

AgriConnect & AgriMind AI

Full-Stack Farm-to-Consumer Marketplace with AI Disease Diagnosis and Fallback Geolocation

1. Project Executive Summary

AgriConnect is a comprehensive digital marketplace designed to bridge the gap between local farmers and consumers. By bypassing traditional intermediaries, the platform ensures fair compensation for producers and access to fresh, organic goods for buyers. Additionally, it integrates advanced AI tools for crop disease diagnosis and automated farm advisory, alongside high-accuracy GPS tracking and geocoding to streamline the delivery logistics.

2. Core Technology Stack

- **Frontend Application:** React.js, Vite build tool, TypeScript, Tailwind CSS styling
- **Backend Services:** Node.js runtime, Express.js web server framework, TypeScript compile
- **Database Engine:** SQLite database engine with active connection state tracking
- **AI / ML Modules:** TensorFlow & Keras (disease model), Google Gemini AI API (advisor chat)
- **Mapping Libraries:** Google Maps JavaScript API, Leaflet maps, OpenStreetMap Nominatim Geocoding

3. Core Feature Specifications

A. AI Crop Disease Diagnosis

Allows farmers to upload crop leaf images. A convolutional neural network (CNN) model classifies diseases (e.g. Late Blight vs Early Blight) and offers immediate treatment advice.

B. AI Agricultural Advisory Chat

A conversation interface powered by Gemini API that answers farmer queries about crop rotation, pesticides, water management, and soil health with localized farming recommendations.

C. Quantity-Based Decimal Ordering

Customers can order custom decimal stock sizes (e.g. 5.5 kg instead of fixed units). The system automatically computes live stock valuations, estimates costs, and updates quantities inside the database.

D. Hybrid Geolocation Maps with CDN Fallback

Requests user GPS with high accuracy. If a Google Maps API Key is active, it renders Google Maps. If missing, it falls back to Leaflet and OpenStreetMap. Auto-populates inputs using reverse geocoding.

E. Multi-Step Checkout Wizard

Refactors the lengthy checkout form into a compact 3-step modal (Contact details -> Delivery Location -> Confirm details) reducing visual clutter and checking validations before moving forward.

F. Live Delivery Tracking Map & Geolocation Fallbacks

Renders an interactive Leaflet OpenStreetMap view inside the GoogleMapComponent when the Google Maps API key is missing. This replaces the static SVG route placeholder. Enables delivery partner location sharing with high-accuracy browser geolocation updates (watchPosition) and pushes position, accuracy, and timestamp telemetry to the server every few seconds. Custom map pins represent Farmer, Customer, and Delivery Partner.

G. Webcam & Photo Capturing for Products

Allows farmers listing vegetables in the marketplace to capture live photos using their webcam stream in real time, or upload files from local storage. Handles immediate uploads and displays live thumbnail previews.

4. User Portal & Role Matrix

1. Customer Portal

Browse fresh catalogs, add custom stock quantities, utilize live GPS geolocation checkouts, track order deliveries, receive fulfillment OTP codes, and file claims/disputes on damaged orders.

2. Farmer Dashboard

List fresh items (specifying decimals, unit dropdowns, and dynamic valuations), view order logs, track accepting deadlines, dispatch shipments, review AI diagnostics, and read AI advisors.

3. Delivery Partner Panel

Complete self-onboarding, claim order shipments, review navigation routes on map indicators, and complete deliveries by entering secure OTP codes supplied by customer buyers.

4. System Administrator Portal

Monitor overall platform stats, run the transaction order explorer ledger, manage dispute resolutions (REFUND or RELEASE funds), and browse the active products directory.